

Solvency Field Theory V12.11

Master Lagrangian v1.1: Sigma2/Full-Current Support Lock and Adaptive Saturation Window

Shaun Fosmark
SFT Research Notes

R025 h-lock update generated: 2026-06-09T05:25:57Z

Abstract

SFT V12.11 updates the original repaired SFT Lagrangian into a single Master Lagrangian document. The repaired V11 action is preserved unchanged as the macro ledger core, S_{core} . The post-V11 developments through V12.10 are not forced into that core term. Instead, V12.11 gives them explicit placement in an action stack: maintained worldline-history sources, no-overwrite admissibility, support-domain constraints, frame/holonomy interface structure, effective mass-focusing and formation layers, and a weak-field source-geometry projection bridge.

The main conclusion is deliberately bounded. V12.11 does not claim a completed final microphysical action, a particle spectrum map, a gauge-coupling derivation, a SPARC re-fit, physical masses, or physical abundances. It claims a tested Master Lagrangian v1 placement map: which sectors are action-ready, which are constraint-level, which are interface-ready, which are effective-only, and which remain open.

The Master Action: Front-Center Statement

The V12.11/R025 construction is centered on the following stacked action. The first line preserves the repaired V11 core. The later lines place the V12.11 upgrades: maintained worldline histories, no-overwrite/support constraints, frame/holonomy interface structure, effective mass and thermal sectors, and the weak-field projection bridge. R025 updates the smooth lock scalar to the sigma2/full-current form earned in R024B.

$$\begin{aligned} S_{\text{SFT}}^{12.11,v1} = & \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \eta \nabla_\mu \chi J_{\text{ops}}^\mu + \Lambda_C C_{\text{min}} - V_{\text{quad}}(\psi) + \mathcal{L}_m - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{\kappa_\theta}{2} D_\mu \theta D^\mu \theta \right] \\ & + \sum_i \int d\tau_i \mathcal{L}_{\text{hist}}[X_i^\mu, \phi_i, \nu_{C,i}] \\ & + \int_\Sigma d^3x \sqrt{\gamma} \left[\Lambda_{\text{NOW}} C_{\text{NOW}} + \Lambda_{\text{support}} C_{\text{RC}} + \Lambda_{\text{cell}} \left(\Pi_{\text{adm}}(\text{cell}) - \frac{3}{4} \right) \right] \\ & + S_{\text{focus}}^{\text{eff}} + S_{\text{eff/thermal}} + \mathcal{P}_{\text{weak-field}}[J_{\text{ops}}, M_{ab}, h, \Pi_{\text{adm}}, \lambda]. \end{aligned} \tag{1}$$

The required definitions are:

$$J_{\text{ops}}^\mu(x) = \sum_i \int d\tau_i \nu_{C,i} u_i^\mu(\tau_i) \delta_g^{(4)}(x - X_i(\tau_i)), \quad (2)$$

$$\nu_{C,i} = \frac{m_i c^2}{h}, \quad (3)$$

$$\Pi_{\text{adm}}(x) = 1 - \frac{h_{\text{lock}}(x)}{4}, \quad (4)$$

$$h_{\text{lock}}(x) = h_{\text{planar}}(x) [1 - f_\perp(x)], \quad (5)$$

$$h_{\text{planar}}(x) = 1 - \sqrt{\frac{\sigma_3^2(x)}{\sigma_2^2(x)}}, \quad h_{\text{planar}} = 0 \text{ when } \sigma_2^2 \rightarrow 0, \quad (6)$$

$$f_\perp(x) = \frac{\langle (J_{\text{spatial}} \cdot n)^2 \rangle_x}{\langle |J_{\text{spatial}}|^2 \rangle_x}, \quad (7)$$

$$\lambda_{\text{disk}} = \left(\frac{\pi}{2}\right) \left(\frac{3}{4}\right) = \frac{3\pi}{8}. \quad (8)$$

This front-center equation is the single-page answer to what the V12.11 Lagrangian is. The rest of the document explains why each term is kept, added, downgraded to effective status, or left open.

Contents

The Master Action: Front-Center Statement	1
1 Canonical Decision	4
2 Claim Boundary	4
3 The Master Lagrangian v1	4
4 Repaired Core Action	5
5 Why V12.11 Extends the Original Lagrangian	5
6 Worldline-History Source Bridge	6
7 Helical Clock Interpretation	6
8 Source-Force Boundary Result	6
9 Motion Through Settlement Geometry	7
10 No-Overwrite Admissibility Sector	7
11 Support-Cell Projector	7
12 Disk Cell Derivation of 3/4	8
13 Smooth Lock Functional: R025 Update	8
14 Canonical Support-Domain Constraint	10

15 Frame / Holonomy Interface	10
16 Weak-Field Projection Bridge	10
17 Disk Endpoint	11
18 Multi-Component Projection Candidate	11
19 Effective Mass-Focusing Sector	11
20 Effective Formation / Thermal Sector	12
21 Master Lagrangian v1 Summary	12
22 Open Ledger	12
23 Checkpoint Verdict	13
A Run Ledger	13
B Null-Test Rule	14
C R024/R024B Support-Lock Audit	14

1 Canonical Decision

The earlier SFT Lagrangian work was split across a focused action document and derivation/history companions. V12.11 is written as a single main document instead.

This document keeps the advantage of the split by separating the content internally:

1. the master action statement,
2. the source-history and no-overwrite upgrades,
3. the weak-field projection bridge,
4. the derivation/audit notes,
5. the open ledger.

The reason for one canonical document is synchronization. V12.11 changes the interpretation of the source ontology. The core action remains valid, but the theory now requires an explicit action stack around it. One document avoids maintaining separate action and derivation files that can drift apart.

2 Claim Boundary

V12.11 claims:

A Master Lagrangian v1 stack that preserves the repaired V11 action as S_{core} , while placing V12.4–V12.10 structures in explicit source-history, no-overwrite, support, frame, effective, and weak-field projection sectors.

V12.11 does not claim:

- final continuum Euler–Lagrange closure;
- a full galaxy morphology solver;
- a SPARC re-fit;
- a derivation of α , gauge couplings, weak mixing, or charge quantization;
- a physical particle spectrum map;
- physical particle masses;
- physical abundance, BBN, CMB, or Boltzmann cosmogenesis closure.

3 The Master Lagrangian v1

The current V12.11 stack is:

$$S_{\text{SFT}}^{12.11,v1} = S_{\text{core}} + S_{\text{hist}} + S_{\text{NOW}} + S_{\text{frame}} + S_{\text{support}} + S_{\text{focus}}^{\text{eff}} + S_{\text{eff/thermal}} + S_{\text{shape}/\lambda} \quad (9)$$

The sectors are:

Sector	Status	Role
S_{core}	native action core	repaired V11 macro ledger action
S_{hist}	bridge-ready	maintained worldline histories to J_{ops}
S_{NOW}	constraint layer	no-overwrite / history-lock admissibility
S_{frame}	interface-ready	framing, holonomy, gauge-compatible interface
S_{support}	canonical constraint	support-domain and support-cell rules
$S_{\text{focus}}^{\text{eff}}$	effective candidate	mass-focusing placement
$S_{\text{eff/thermal}}$	effective toy	formation/racetrack kinetic layer
$S_{\text{shape}/\lambda}$	weak-field bridge	source geometry to morphology coefficient

Equation (9) is not yet a final unified action. It is a placement map. The essential V12.11 update is that the original Lagrangian is no longer asked to carry every later V12 structure inside a single scalar source term. The repaired core remains the macro action, while the added V12 sectors specify how the source itself is now understood.

4 Repaired Core Action

The repaired V11 action is preserved:

$$S_{\text{core}} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \eta \nabla_\mu \chi J_{\text{ops}}^\mu + \Lambda_C C_{\text{min}} - V_{\text{quad}}(\psi) + \mathcal{L}_m \right]. \quad (10)$$

Here:

- $R/(16\pi G)$ is the Einstein–Hilbert sector;
- $\chi = \bar{\chi} + \psi$ is the ledger/bookkeeping field split into background and lag fluctuation;
- J_{ops}^μ is the operation-density current;
- C_{min} is the count-normalization constraint;
- $V_{\text{quad}}(\psi)$ is the repaired background-relative lag potential;
- $\mathcal{L}_m$ is the matter sector.

The repaired lag potential satisfies:

$$V_{\text{quad}}(0) = 0, \quad V'_{\text{quad}}(0) = 0, \quad V''_{\text{quad}}(0) > 0. \quad (11)$$

Thus, on the matched static vacuum branch, the non-Einstein bookkeeping sectors vanish and S_{core} reduces to the Einstein–Hilbert branch. This keeps exact exterior GR recovery available while preserving nontrivial ledger dynamics away from the matched vacuum branch.

5 Why V12.11 Extends the Original Lagrangian

The original repaired Lagrangian treated matter primarily as a source current:

$$\text{matter clocks} \rightarrow J_{\text{ops}}^\mu \rightarrow S_{\text{core}}.$$

The current V12 ontology is deeper:

maintained no-overwrite worldline histories $\rightarrow$ cadence/framing/support/focusing $\rightarrow J_{\text{ops}}^\mu \rightarrow S_{\text{core}}$.

This is the essential reason for V12.11. The old action is not discarded. It is reclassified as the macro ledger core. The V12 upgrade asks what the action stack must know about the particle histories that generate the source current.

6 Worldline-History Source Bridge

V12.11 defines the operation current as the coarse-grained current of maintained worldline histories:

$$J_{\text{ops}}^\mu(x) = \sum_i \int d\tau_i \nu_{C,i} u_i^\mu(\tau_i) \delta_g^{(4)}(x - X_i(\tau_i)). \quad (12)$$

The Compton cadence is:

$$\nu_C = \frac{mc^2}{h}. \quad (13)$$

This bridge does not by itself derive the particle spectrum. It says that the current used by the repaired macro action can be read as the continuum shadow of maintained particle histories.

7 Helical Clock Interpretation

The V12.11 particle language is:

Particles are maintained helical worldline structures in $3 + 1$ spacetime.

The word “helical” is not a preferred-frame corkscrew through space. It denotes internal Compton phase / normal-frame holonomy carried along a worldline:

$$\phi_i(\tau) = \omega_{C,i}\tau, \quad \omega_{C,i} = 2\pi\nu_{C,i}. \quad (14)$$

The observed particle is a local slice/projection/update-front of a maintained spacetime history. The history already written is fixed. Only the next compatible continuation is variationally open.

8 Source-Force Boundary Result

Substituting the worldline current into the direct χ -current coupling gives:

$$S_{\chi,i} = -\eta\nu_{C,i} \int d\tau u_i^\mu \partial_\mu \chi(X_i(\tau)) \quad (15)$$

$$= -\eta\nu_{C,i} \int d\tau \frac{d\chi(X_i(\tau))}{d\tau} \quad (16)$$

$$= -\eta\nu_{C,i} [\chi(X_f) - \chi(X_i)]. \quad (17)$$

Therefore, for conserved fixed-cadence histories:

$$-\eta\nabla_\mu \chi J_{\text{ops}}^\mu \text{ is source/bookkeeping boundary structure, not a raw scalar fifth-force.} \quad (18)$$

This is a stabilizing result. It prevents the repaired core from producing an unwanted local χ -gradient push on conserved particles.

9 Motion Through Settlement Geometry

After Eq. (17), local motion is placed in the metric/settlement geometry channel:

J_{ops} source geometry $\rightarrow S_{\text{core}} \rightarrow$ settlement / metric geometry $\rightarrow$ future-write trajectory.

In the weak-field limit, ordinary geometric motion has the schematic form:

$$\frac{d^2 \mathbf{x}}{dt^2} = -\nabla \Phi. \quad (19)$$

The SFT distinction is interpretive and structural: Φ is not sourced by a raw fifth force from χ , but through the settlement/metric branch generated by the ledger source.

10 No-Overwrite Admissibility Sector

The no-overwrite principle is:

Previously written support is fixed. Dynamics can vary only the next compatible write.

V12.11 places this principle in a constraint/admissibility sector:

$$S_{\text{NOW}} = \int_{\Sigma} d^3x \sqrt{\gamma} \Lambda_{\text{NOW}}(x) C_{\text{NOW}}(x), \quad (20)$$

where $C_{\text{NOW}} = 0$ enforces append-only support compatibility.

The exact continuum form remains open. V12.11 supplies the tested support-cell form and a smooth projector target.

11 Support-Cell Projector

Define a local support-cell admissibility projector:

$$\Pi_{\text{adm}}(\text{cell}) = \frac{1}{N_{\text{cell}}} \sum_s A_s, \quad A_s \in \{0, 1\}. \quad (21)$$

Here $A_s = 1$ if local state s is admissible, and $A_s = 0$ if local state s is forbidden by no-overwrite/support compatibility.

The constraint contribution is:

$$S_{\text{NOW/support}}^{\text{cell}} = \sum_{\text{cells}} \Lambda_{\text{cell}} C_{\text{cell}}. \quad (22)$$

For a planar disk support cell:

$$C_{\text{NOW,disk}} := \Pi_{\text{adm}}(\text{cell}) - \frac{3}{4} = 0. \quad (23)$$

12 Disk Cell Derivation of 3/4

The planar disk support cell has four 3 + 1 slots:

Slot	Status	Reason
planar tangent x	admissible	in-plane spatial support
planar tangent y	admissible	in-plane spatial support
forward write	admissible	compatible next-write continuation
normal/history lock	forbidden	would overwrite locked normal/history support

Therefore:

$$\boxed{\Pi_{\text{adm,disk}} = \frac{3}{4}.}$$
 (24)

The same count appears in the binary no-overwrite cell:

support channel $\in \{\text{planar support, normal/history support}\},$

write status $\in \{\text{compatible next write, overwrite locked history}\}.$

There are four states. Exactly one is forbidden:

normal/history support + overwrite locked history.

Thus:

$$\Pi_{\text{adm}} = \frac{3}{4}.$$

13 Smooth Lock Functional: R025 Update

The support-cell rule can be smoothed by introducing a history-lock scalar:

$$h_{\text{lock}}(x) \in [0, 1].$$
 (25)

The admissibility projector is now:

$$\boxed{\Pi_{\text{adm}}(x) = 1 - \frac{h_{\text{lock}}(x)}{4}.}$$
 (26)

Thus:

$$h_{\text{lock}} = 0 \Rightarrow \Pi_{\text{adm}} = 1, \quad h_{\text{lock}} = 1 \Rightarrow \Pi_{\text{adm}} = \frac{3}{4}.$$

R024B replaces the earlier pure-shape candidate with an upstream no-overwrite support-lock scalar. Define a local written-support tensor:

$$W_{ab}^{(\ell)}(x) = \frac{\int_{\mathcal{N}_\ell(x)} d^3y K_\ell(x, y) \rho_{\text{ops}}(y) (y_a - \bar{y}_a)(y_b - \bar{y}_b)}{\int_{\mathcal{N}_\ell(x)} d^3y K_\ell(x, y) \rho_{\text{ops}}(y)}.$$
 (27)

Let its ordered eigenvalues be:

$$\sigma_1^2(x) \geq \sigma_2^2(x) \geq \sigma_3^2(x) \geq 0.$$

The local normal $n(x)$ is the eigenvector associated with σ_3^2 .

The planar support-lock scalar is:

$$h_{\text{planar}}(x) = 1 - \sqrt{\frac{\sigma_3^2(x)}{\sigma_2^2(x)}}, \quad h_{\text{planar}} = 0 \text{ when } \sigma_2^2 \rightarrow 0. \quad (28)$$

This is the slot-table scalar. It measures whether the smallest support direction has collapsed relative to the second support direction. It does not require in-plane roundness. Therefore a planar bar can carry full normal-history lock, while a cigar or line-like support cannot.

The history/current correction is:

$$f_{\perp}(x) = \frac{\langle (J_{\text{spatial}} \cdot n)^2 \rangle_x}{\langle |J_{\text{spatial}}|^2 \rangle_x}. \quad (29)$$

The final R025 support-lock scalar is:

$$h_{\text{lock}}(x) = h_{\text{planar}}(x) [1 - f_{\perp}(x)]. \quad (30)$$

This distinguishes a quasi-static disk from a transient pancake. A disk has worldline/current flow primarily inside the plane, so $f_{\perp} \simeq 0$. A transient pancake can be snapshot-flat while worldlines cross the normal direction, so $f_{\perp} \simeq 1$ and the lock vanishes.

For Gaussian ellipsoidal support with axis ratio $r = \sigma_3/\sigma_2$, Eq. (28) gives:

$$h_{\text{planar}} = 1 - r. \quad (31)$$

Thus h has a direct physical reading: it is the fraction of the normal direction written shut. A five-percent-thickness disk has $h \simeq 0.95$, and

$$\Pi_{\text{adm}} = 1 - \frac{0.95}{4} = 0.7625.$$

The 3/4 endpoint is the zero-thickness limit where the normal direction is fully written shut.

Adaptive saturation window

The neighborhood scale ℓ in Eq. (27) is not left as an analyst-chosen parameter. R025 uses the saturation radius:

$$\ell_*(x) = \min \left\{ \ell : \left| \frac{dh_{\text{lock}}^{(\ell)}(x)}{d \ln \ell} \right| < \varepsilon_{\text{sat}} \text{ over a stable plateau} \right\}. \quad (32)$$

Small windows see local nearly isotropic support and correctly return weak lock. Once the window resolves the written support structure, $h_{\text{lock}}^{(\ell)}$ saturates. The structure therefore selects its own support window. The numerical tolerance ε_{sat} is a convergence criterion, not a physical fit parameter.

The earlier R024 scalar

$$q_{\text{flat}}^{\text{Coblate}}$$

is not discarded. It is reclassified as a downstream morphology-purity or oblate-coherence correction for weak-field phenomenology, not the core $S_{\text{NOW}}/S_{\text{support}}$ support-lock scalar.

14 Canonical Support-Domain Constraint

The canonical V12 support seed is:

$$D_{\text{seed}} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, 1, \frac{5}{4}, \frac{4}{3}, \frac{3}{2}, \frac{5}{3}, 2 \right\}. \quad (33)$$

A bare denominator cutoff is insufficient because it admits an unwanted extra mode $7/4$. The canonical rule requires:

1. primitive-anchor support-domain membership generated from $\{1, 1/2, 1/3\}$,
2. binary no-overwrite distinguishability,
3. $3 + 1$ slot capacity.

This places the nine-mode seed in S_{support} , not as a completed physical particle spectrum.

15 Frame / Holonomy Interface

The frame sector remains interface-ready:

$$D_\mu \theta = \partial_\mu \theta - q A_\mu. \quad (34)$$

Gauge-compatible transformations are:

$$\theta \rightarrow \theta + q\lambda, \quad A_\mu \rightarrow A_\mu + \partial_\mu \lambda, \quad F = dA \text{ invariant}. \quad (35)$$

This supports the V12.4 reading of charge as worldline framing / boundary-address holonomy, but does not derive α , charge quantization, weak mixing, Yukawas, CKM/PMNS, or particle identities.

16 Weak-Field Projection Bridge

The weak-field branch is:

$$S_{\text{core}} \rightarrow J_{\text{ops}} \text{ source geometry} \rightarrow \text{raw weak-field projection} \rightarrow S_{\text{NOW}}/S_{\text{support}} \text{ admissibility} \rightarrow \lambda.$$

The source-shape moment tensor is:

$$M_{ab}[J_{\text{ops}}] = \frac{\int d^3x \rho_{\text{ops}}(x)(x_a - \bar{x}_a)(x_b - \bar{x}_b)}{\int d^3x \rho_{\text{ops}}(x)}. \quad (36)$$

For eigenvalues:

$$\sigma_1^2 \geq \sigma_2^2 \geq \sigma_3^2,$$

define:

$$q_{\text{flat}} = 1 - \sqrt{\frac{\sigma_3^2}{(\sigma_1^2 + \sigma_2^2)/2}}. \quad (37)$$

R025 separation: support lock versus morphology purity

The q_{flat} scalar remains useful as a weak-field morphology/readout quantity, but R024B shows that it is not the clean upstream no-overwrite lock scalar. The lock scalar belongs to $S_{\text{NOW}}/S_{\text{support}}$ and is:

$$h_{\text{lock}} = \left(1 - \sqrt{\frac{\sigma_3^2}{\sigma_2^2}}\right) (1 - f_{\perp}).$$

The $q_{\text{flat}}c_{\text{oblate}}$ combination is therefore demoted to an optional downstream morphology-purity/coherence factor. This matters for planar bars: the slot table says a planar bar can fully lock the normal/history slot, but the weak-field morphology bridge may still treat barred structures as phenomenologically different from clean axisymmetric disks.

The first morphology bridge is:

$$\boxed{\lambda = \lambda_{\text{sphere}} - (\lambda_{\text{sphere}} - \lambda_{\text{disk}})q_{\text{flat}}.} \quad (38)$$

The endpoints are:

$$\lambda_{\text{sphere}} = \frac{4}{3}, \quad \lambda_{\text{disk}} = \frac{3\pi}{8}. \quad (39)$$

17 Disk Endpoint

The raw ideal disk chord average is:

$$\text{raw}_{\text{disk}} = \frac{\pi}{2}. \quad (40)$$

The V12 support-cell admissibility factor is:

$$\Pi_{\text{adm,disk}} = \frac{3}{4}.$$

Therefore:

$$\boxed{\lambda_{\text{disk}} = \text{raw}_{\text{disk}} \Pi_{\text{adm,disk}} = \left(\frac{\pi}{2}\right) \left(\frac{3}{4}\right) = \frac{3\pi}{8}.} \quad (41)$$

This is a constraint-promoted weak-field bridge, not a fitted scalar force.

18 Multi-Component Projection Candidate

A candidate multi-component extension is:

$$\lambda_{\text{total}} = \sum_j w_j \lambda_j(\text{type}_j, M_j), \quad \sum_j w_j = 1. \quad (42)$$

This prevents bars, warps, rings, finite-thickness disks, and bulges from being silently treated as ideal thin disks. Current tests are synthetic stress tests only. No SPARC re-fit is claimed.

19 Effective Mass-Focusing Sector

The V12.9 mass sector remains:

$$S_{\text{focus}}^{\text{eff}} \quad \text{effective/candidate only.} \quad (43)$$

A schematic placement is:

$$M_{\text{structured}} = M_{\text{bridge}}(\rho) G_{\text{shape}}(m, N) Q_{\text{force}}(\text{residue, topology, ecology}). \quad (44)$$

V12.11 does not promote this to a native covariant mass derivation.

20 Effective Formation / Thermal Sector

The V12.10 formation racetrack remains:

$$S_{\text{eff/thermal}} \quad \text{effective kinetic layer only.} \quad (45)$$

A schematic kinetic form is:

$$\dot{Y}_i \approx \Gamma_{\text{form},i} - \Gamma_{\text{disrupt},i} - 3HY_i - \Gamma_{\text{decay},i}Y_i. \quad (46)$$

This keeps heavy-first / light-last sorting and formation/disruption structure, but does not claim physical abundances, BBN yields, CMB spectra, or Boltzmann cosmogenesis closure.

21 Master Lagrangian v1 Summary

The single-document V12.11 Master Lagrangian is:

$$\boxed{S_{\text{SFT}}^{12.11,v1} = S_{\text{core}} + S_{\text{hist}} + S_{\text{NOW}} + S_{\text{frame}} + S_{\text{support}} + S_{\text{focus}}^{\text{eff}} + S_{\text{eff/thermal}} + S_{\text{shape}/\lambda}.} \quad (47)$$

With:

$$S_{\text{core}} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \eta \nabla_\mu \chi J_{\text{ops}}^\mu + \Lambda_C C_{\text{min}} - V_{\text{quad}}(\psi) + \mathcal{L}_m \right], \quad (48)$$

$$J_{\text{ops}}^\mu(x) = \sum_i \int d\tau_i \nu_{C,i} u_i^\mu(\tau_i) \delta_g^{(4)}(x - X_i(\tau_i)), \quad (49)$$

$$S_{\text{NOW/support}}^{\text{cell}} = \sum_{\text{cells}} \Lambda_{\text{cell}} C_{\text{cell}}, \quad (50)$$

$$C_{\text{NOW,disk}} = \Pi_{\text{adm}}(\text{cell}) - \frac{3}{4} = 0, \quad (51)$$

$$\Pi_{\text{adm}}(x) = 1 - \frac{h_{\text{lock}}(x)}{4}, \quad (52)$$

$$h_{\text{lock}}(x) = \left[1 - \sqrt{\frac{\sigma_3^2(x)}{\sigma_2^2(x)}} \right] [1 - f_\perp(x)], \quad (53)$$

$$f_\perp(x) = \frac{\langle (J_{\text{spatial}} \cdot n)^2 \rangle_x}{\langle |J_{\text{spatial}}|^2 \rangle_x}, \quad (54)$$

$$\lambda_{\text{disk}} = \left(\frac{\pi}{2} \right) \left(\frac{3}{4} \right) = \frac{3\pi}{8}. \quad (55)$$

22 Open Ledger

The open problems after V12.11 are:

1. derive the R025 $h_{\text{lock}}(x)$ fully from metric/source/support field equations rather than from the current tensor candidate;
2. convert $S_{\text{NOW}}/S_{\text{support}}$, including the adaptive saturation window ℓ_* , into a fully smooth continuum variational sector;
3. derive the full weak-field projection operator beyond synthetic component tests;

4. derive or constrain gauge couplings and α ;
5. map support-domain modes, if possible, to physical sectors without overclaiming;
6. promote mass focusing beyond effective/candidate status;
7. promote formation racetrack beyond toy kinetic status;
8. connect the weak-field projection bridge to a full galaxy solver if data are later used.

23 Checkpoint Verdict

SFT V12.11 is ready as the canonical Master Lagrangian checkpoint.

The repaired V11 action survives unchanged as S_{core} . The later V12 machinery now has explicit formal placement. The source ontology is upgraded from primitive ticking matter to maintained no-overwrite worldline histories whose coarse-grained current sources the macro ledger action.

The document is intentionally bounded:

action-ready core + tested constraints + bridge-ready sectors + effective layers + open ledger.

This is the right status for V12.11.

A Run Ledger

Run	Status	Result
R001	pass	action audit matrix
R002	pass	repaired core survives
R003	pass	worldline current bridge
R004	pass	no-overwrite admissibility toy model
R005	pass	helical clock/phase compatibility
R006B	pass after patch	canonical support-domain constraint; 7/4 excluded
R007	pass	frame/holonomy gauge interface
R008	pass effective	mass focusing placed as effective/candidate
R009	pass effective	formation racetrack placed as effective kinetic layer
R010–R011	pass	Master Action stack and equation skeleton
R012	boundary result	direct $\nabla\chi \cdot J$ is boundary/source bookkeeping
R013	pass	motion placed in metric/settlement geometry
R014–R019	pass	weak-field λ bridge and disk 3/4 constraint promotion
R020	pass	Master Action v1 rollup
R021–R023	pass with patches	formal note, smooth projector, $h(x)$ candidate, P-shape stress tests, readiness gate

B Null-Test Rule

SFT V12.11 distinguishes two kinds of null tests:

1. desired invariance nulls, such as lane/order/rotation/scale changes, where a structural result should survive;
2. specificity nulls, such as arbitrary label shuffles, where success would indicate triviality.

The runner analogy is:

If the fast runner wins no matter the lane, that is evidence of structural inevitability. If random bib numbers predict the winner, the test is broken.

This rule is preserved in the weak-field morphology bridge.

C R024/R024B Support-Lock Audit

R024 first tested:

$$h_{\text{R024}} = q_{\text{flat}}(W_{ab})c_{\text{oblate}}(W_{ab}).$$

That scalar passed endpoint and hygiene tests and suppressed bar/prolate false positives. The R024B addendum showed that it was answering a downstream morphology-purity question, not the pure slot-table lock question.

The slot-table lock question is different. The normal/history slot is locked when support has become planar, regardless of whether the in-plane distribution is round or elongated. Therefore a planar bar can carry full normal-history lock. The winning upstream scalar is:

$$h_{\text{planar}} = 1 - \sqrt{\frac{\sigma_3^2}{\sigma_2^2}},$$

with $h_{\text{planar}} = 0$ when $\sigma_2 \rightarrow 0$.

R024B also added the full-current correction:

$$h_{\text{lock}} = h_{\text{planar}}(1 - f_{\perp}).$$

This passed the transient-pancake null: a snapshot-flat but normally crossing configuration has $h_{\text{planar}} \simeq 1$, $f_{\perp} \simeq 1$, and $h_{\text{lock}} \simeq 0$.

The audit ruling is:

Use $h_{\text{lock}} = [1 - \sqrt{\sigma_3^2/\sigma_2^2}](1 - f_{\perp})$ for $S_{\text{NOW}}/S_{\text{support}}$. Preserve $q_{\text{flat}}c_{\text{oblate}}$ only as an optional downstream morphology-purity correction.

The remaining soft flags are:

1. a complete field-equation derivation of W_{ab} , $f_{\perp}$, and ℓ_* is still open;
2. morphology-purity corrections for real barred, warped, gas-rich, or multi-component galaxies remain downstream weak-field work, not part of the core support-lock scalar.